28 For an electron passing through a single slit, the uncertainty in its position Δy and the uncertainty in its momentum Δp are related by

$$\Delta y \, \Delta p \gtrsim h.$$

Which diagram shows the positions where Δy and Δp in the equation above are defined and shows them in the correct direction?

A



B



C



D



L2 **Answer: A**

In the setup, the **electrons** passing through the slit will exhibit **wave nature** at the beginning of the setup (i.e. the slit). **Uncertainties will apply to the position at the slit.**

Upon passing through the slit, the electrons spread out in the *y* direction (up/down). Hence uncertainty in their momentum will also be in *y* direction.
It is the **p and y which are along the same direction** that are **paired** in Heisenberg Uncertainty Principle.

